SWEEP rule on the KD-Toric code

February 18, 2026

1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+. If there are more than one, then pick the one which has the minimal weight.

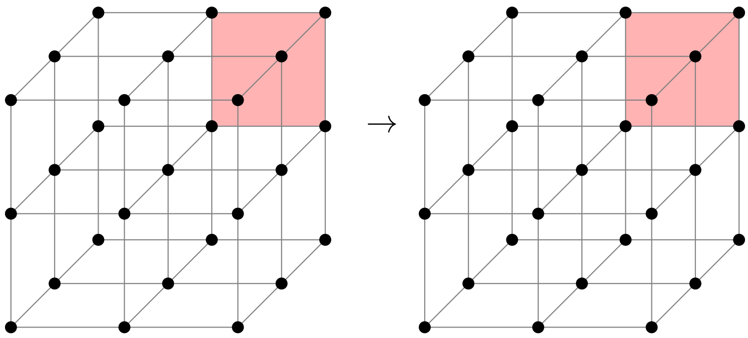

Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow[1]:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t, and let H_e be the unsatisfied checks at time $t-1$. Then there is a set $H'_e \subset H_e$ such that:

[1] Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. For any $i < d+1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.

2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. Consider a vertex v which at time t adjoins to an unsatisfied edge e, let u, w be vertices which have a positive faces that adjoin to e.

1. If $\hat{u}|_e \neq \emptyset$, then the parallel edge to e going out from u belongs to H_e. Denote it by e_u, taking e_u satisfies at least two (three in 4D) of the first d inequalities and the last $d+1$th inequality. Denote by $x \in [d]$ the index of the inequality which isn't satisfied by e_u. So it remains to prove the existence of $e' \in H_e$ for which $L_x(e') \geq L_x(e)$.

   Consider the assignment of bits at time $t-1$ over the complex, and consider the graph $G = (\mathcal{F}, E)$ induced by associating vertices with any face that is set to 1 by the assignment. Two vertices are connected if their origin faces share an edge. Let $K \subset \mathcal{F}$ be a connected component of faces that contains the faces supported on u.

   We say that K is d far from u if:

$$\max_{f \in K} |f_x - u_x| \leq d$$

   Now, we will prove by induction on d that there is a face $f \in K$ with $f_x \geq v_x$ such one of its edge is not satisfied.

   (a) **Base.** The edge from v which point at the $\hat{z}$ direction must not be satisfied.

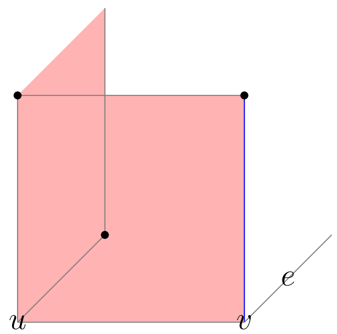

Figure 2: The induction base: In red, we have the faces on the support of u. The blue edge has the same x-coordinate as e. Any attempt to zero out the syndrome on the edge results in adding a face that is at least 2-far from u. Any configuration which is at most 1-far from u has to have that local view, yet might be connected to more faces which don't adjoin the blue edge.

   (b) **Assumption.** Assume that for any K which is at most $d-1$ far from u there is a face, with adjoins edge that is both has a x-coordinate that is greater than u_x and is not satisfied.

(c) **Induction Step.** Consider a connected component K that is d-far from u. Notice that if by substitute a co-boundary (stabilizer) from K we get K' that is at most d' far from u for some $d' < d$ then we done. That because: $\partial_2 K' = \partial_2(K+z) = \partial_2 K$ for some $z \in \ker \partial_2$.

On the other hand, **[COMMENT]** Finish that.

2. Else, if for any $u \in \downarrow(u) \cap lk_1$ it holds that $\hat{u}|_e = \emptyset$, then it must be that $e \in H_e$ because $\hat{v}$ can only decrease the syndrome on v. Moreover, it follows that $\hat{v} = \emptyset$ (otherwise, the suggestion would zero out the syndrome on $\uparrow(v) \cap lk_2$). Thus, $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$ there is also an edge $e' \in H_e$ where $e' \in \downarrow(v)$. In that case, take $e \in H_e$ as the edge satisfies the first d equalities and e' for satisfying the $d+1$th inequality.

$\square$

3 Investigation

[COMMENT] To do: understands how the investigation should look like. Plot an example to such tree. The model goes as follow, at each even iteration we toss a noise, then in each odd iteration we perform a single SWEEP rule iteration. That allows to track over the history of the errors.

We say that a check $e \in$ Error, if there is a face f that adjoin to e and belongs to the Error.

Definition 3.1 (Investigation)**.** For any unsatisfied check e at time t, we define V (the set of checks which participated in 'misleading' of e), and two sets of edges on V, Arrows and Forks via the following algorithm:

let $V = \{e\}$, Arrows $= \emptyset$, Forks $= \emptyset$
let $Q \leftarrow$ queue $\{e\}$
while *Q is not empty* **do**
    $e' \leftarrow$ dequeue Q
    if $e' \notin$ *Error* **then**
        $e_1, .., e_l \leftarrow$ Excuse(e')
        Insert $\{e_i, e'\}$ to Arrows
        Insert $\{e_i, e_j\}$ to Forks
    end
end

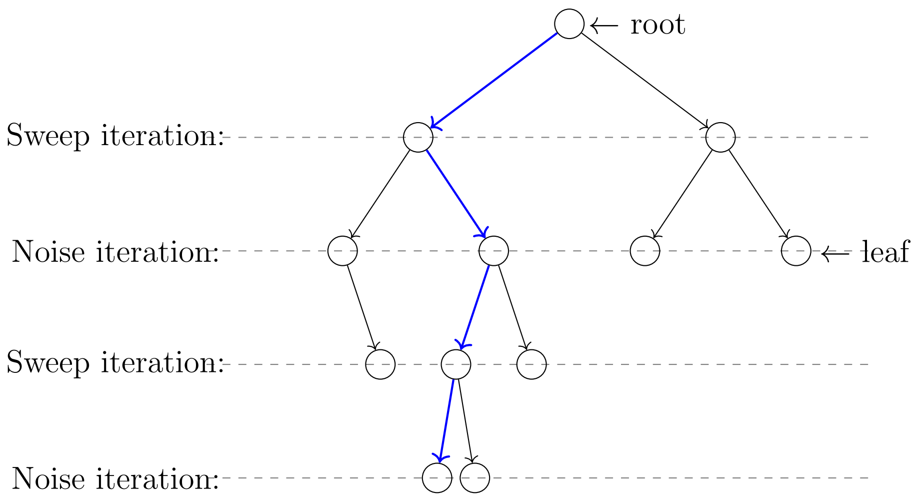

Figure 3: blabla

4 Space-Time Graph

Definition 4.1. Consider a graph $G = (V, A, F)$, where V is a set of vertices, and A, F are sets of undirected edges. For $v \in V$, denote by $\text{pre}(v) = \{u \in V : \{u, v\} \in A\}$, we extend the notion for subsets of vertices $X \subset V$ by $\text{pre}(X) = \bigcup_{v \in X} \text{pre}(v)$. Let $T : V \to \mathbb{N}$, We say that (G, T) is a time graph iff:

1. $\{x, y\} \in A \Rightarrow T(y) = T(x) \pm 1$
2. For $x, y \in V$ $\{x, y\} \in F$ iff there exists $z \in V$ such both $\{x, z\}, \{y, z\} \in A$.

We will call to T time-function, to A arrows, and to F forks. Observes that from the second requirement it follows that forks can connect only vertices that have the same time value (set by T).

Definition 4.2 (History, slice.)**.** For a time graph (G, T) and a vertex $u \in V$, let G_u be the subgraph of (V, A) induced by $\{v \in V : T(v) \leq T(u)\}$. We will denote by H_u the 'History of u', which is the set of vertices that are connected to u in G_u. A slice is an equivalence class of the relation $u \equiv v$ iff $H_u = H_v$. Notice that the time function has to be constant on a slice, to see this consider vertices x, y with $T(x) > T(y)$, thus $x \notin G_y$ and therefore $x \in H_x / H_y$ So they are belong to different classes. Thus, we can extend the time function and the history to slices and write for a slice K: $T(K)$ and H_K.

Claim 4.1. Consider slices K_1, K_2 such that $T(K_1) = T(K_2)$, then either $K_1 = K_2$ or H_{K_1}, H_{K_2} are disjoint.

Proof. Assume a non-empty intersection, so there exists $z \in H_{K_1} \cap H_{K_2}$ such that for any $x \in K_1$ and $y \in K_2$, there exist arrow-paths $z \to x$ and $z \to y$. Thus, for any $w \in H_{K_1}$, one can concatenate a path $w \to x \to z \to y$ and conclude that $y \in H_{K_2}$. $\square$

Definition 4.3 (The graph of slice)**.** Let K be a slice. The graph of K, denoted by $G_K = (V_K, E_K)$, is the graph whose vertices are slices $R \subset H_K$ with $T(R) = T(K) - 1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S, R \in V_K$ are connected in G_K if and only if there is $\{s, r\} = f \in F$ such that $s \in S$ and $r \in R$.

Claim 4.2. G_K is connected.

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider such a path that is minimal in the number of points x belong to it, with $T(x) = T(K)$. By induction on the number k of such points we prove that R and S are connected in G_K.

1. Base. $k = 0$, In this case, the path from r to s lies within $G_r(= G_s)$. Hence $H_r = H_s$, So $R = S$.

2. Induction step. There exists y on the path from r to s such that $T(y) = T(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:

   (a) $\{x, z\} \subset \mathrm{pre}(y) \Rightarrow \{x, z\} \in F$.

   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

   Since the path connecting r, x in H_K has $k-1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

□

Definition 4.4. Let $f : G \hookrightarrow \mathbb{R}^d$ be an embedding that maps each element of $V \cup A \cup F$ to a point in $\mathbb{R}^d$. We say that $\left(G, \mathbb{R}^d\right)$ is a space-time graph iff:

1.

2. The point set by f for an edge is the mean of its endpoints' images; that is, for $e = \{x, y\} \in A \cup F$, we have $f(e) = \frac{1}{2}\left(f(x) + f(y)\right)$.

Claim 4.3. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$
\begin{aligned}
\mathbf{Size}(R_1 \cap R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}
$$

□

Definition 4.5 (Spanned Set)**.** Consider an embedding $f : \star \hookrightarrow \mathbb{R}^d$, a subset of f's domain $S \subset \star$, and $d+1$ points $x_1, \ldots, x_{d+1}$. We say that $\langle S, x_1, \ldots, x_{d+1} \rangle$ is a spanned set of S if for any $i \in [d+1]$ and $s \in S$, it holds that $L(f(s)) \leq L_i(x_i)$. We extend the notation to a spanned slice when S is a slice of some time-graph. Notice that saying that $\langle S, x_1, \ldots, x_{d+1} \rangle$ is a spanned set means $\mathbf{Size}(S) \leq \mathbf{Size}(\{x_1, .., x_{d+1}\})$.

Claim 4.4. Let $K = \langle K, v_1, ..v_{d+1} \rangle$ be a spanned slice of the investigation such that $K \neq H_K$. Then for some k there exist spanned slices $K_0, .., K_k$ and Forks $F_1, .., F_k$ such that:

1. $\mathbf{Excuse}_i(v_i)$ belongs to poles of $K_0, .., K_k$.

2. Introduce an edge between K_i and K_j iff one of the Forks $F_1, .., F_k$ consists of a pole of K_i and a pole of K_j. Then the graph formed from $K_0, .., K_k$ is connected.

3. $\sum_{i=0}^{k} \mathbf{Span}(K_i) + \sum_{i=1}^{k} \mathbf{Span}(F_i) \geq \mathbf{Span}(K) + 1$

Proof. First Notice that K doesn't contain points for which **Error** is true: they would not connected to any other points in H_K via arrows. Thus $\mathbf{Excuse}_i(v_i)$ is defined for $i = 1, .., d+1$. Next consider the graph $G_K = \langle V_K, E_K \rangle$ from **??**. V_K consists of those R for which $T(R)+1 = T(K)$ and which are contained in H_K. Since $\{v_i, \mathbf{Excuse}_i(v_i)\}$ is an arrow $\mathbf{Excuse}_i(v_i)$ belongs to a slice R_i from V_K, for $i = 1, .., d+1$.

Since G_K is connected and its edges are realized by forks, there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\mathbf{Excuse}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $K_0, .., K_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $F_1, \cdots, F_k$ in such way that:

$$\sum_{i=0}^{k} \mathbf{Span}(K_i) + \sum_{i=1}^{k} \mathbf{Span}(F_i) = \sum_{i=1}^{d+1} L_i\left(\mathbf{Excuse}_i(v_i)\right) \geq \mathbf{Span}(K) + 1$$

This will assure (3).

Now we describe the selection of poles. Since we have to span.. □

5 Explantation of a Spanned Slice

[COMMENT] Here we follow the original proof, but change the triminalogy, m will be the number of leaves in the explanation, then in the end we conclude that the number of faults is at least m/Δ. So $|R| \leq f(\frac{m}{\Delta})$.

Definition 5.1. Given sets W of points, R of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $R \cup F$. The sets W, R and F form an explanation of a spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ iff:

1. $W \cup \{v_1, .., v_{d+1}\} \subset U \subset H_K$ and the graph $\langle U, R \cup F \rangle$ is connected.

2. All elements of W are errors.

3. For $m = |W|$ and $s = \mathbf{Span}(K)$ we have $|R| \leq \alpha(m-1) - \beta s$ and $|F| < m - 1$.

Claim 5.1. Every spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ has an explanation.

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. The proof is by induction on h.

1. Base. $h = 0$. Then no arrows can connect points of H_K, Thus H_K, as well as $\mathbf{K}$ consists of a single point u such that $u \in$ Error [COMMENT] Leaves. In that case $U = W = \{u\}$, $m = 1, s = 0$ and $|R|, |F| = 0$ satisfies the required.

2. Induction Step. Now, we know that $K \neq H_K$. Thus, by **??**, we know that for some k there exist spanned slices $K_0, .., K_k$ and forks $F_1, .., F_k$ that satisfy **??**.

   Let $s_i = \mathbf{Span}(K_i)$. By the inductive hypothesis, we know that K_i has an explanation $\mathbf{W}_i, \mathbf{R}_i, \mathbf{F}_i$. For $m_i = |\mathbf{W}_i|$, the sets R_i and F_i have at most $\alpha m_i - \beta s_i$ and $m_i - 1$ leaves. We define:

   (a) $\mathbf{W} = \cup_{i=0}^{k} \mathbf{W}_i$, with $m = \sum_{i=0}^{k} m_i$ leaves. Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j, thus by **??** $\mathbf{W}$ is a union of disjoint set.

   (b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^{k} \mathbf{F}_i$ with at most $k + \sum_{i=0}^{k} m_i - 1 = m - 1$ elements.

   (c) $\mathbf{R} = \{\{v_i, \mathbf{Excuse}_i(v_i)\}\} \cup_{i=0}^{k} \mathbf{R}_i$ with at most

$$d + 1 + \sum_{i=0}^{k} \alpha(m_i - 1) - \beta s_i \leq d + 1 + \alpha m - \alpha(k+1) - \beta\left(s - 1 + k \cdot \mathbf{Span}(\text{fork})\right)$$
$$= \alpha m - \beta s + (d + 1 - (k+1)\alpha + \beta - \beta k \cdot \mathbf{Span}(\text{fork}))$$

   So its enough to pick α, β such that for any $k \geq 1$:

$$d + 1 - (k+1)\alpha + \beta - \beta k f \leq 0$$
$$\Rightarrow \beta = 1, \alpha = (d+1)^3(1+f)$$

   It is easy to see that $\mathbf{W}, \mathbf{R}, \mathbf{F}$ form an explanation of K.

□

6 Counting Subtrees.

Claim 6.1. Let $G = (V, E)$ be a $\Delta + 1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. The number of subtrees rooted at r is at most α^k.

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tress. There is an injection between the subtrees of G, rooted at v, with at most k edges, to the subtrees of T_Δ rooted at v, with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v, with at most k edges to the subtrees of T_2, rooted at v, with at most $k \log \Delta$ edges.

[COMMENT] In the second map, we think on replacing each of the vertex with $\Delta - 1$-leaves binary tree, each leaf is associated with a outgoing edge. .

Thus, it's enough to bound the number of of subtrees in T_2 rooted at v, with at most $k \log \Delta$ edges, that number is less than the number of binary tress with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^{k} \frac{\Delta^{2j}}{(j\log\Delta)^{\frac{3}{2}}\sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi}\log^{\frac{3}{2}}\Delta}\sum_{j=1}^{k}\frac{1}{j^{\frac{3}{2}}} \leq \zeta(\frac{3}{2})\cdot\frac{\Delta^{2k}}{\sqrt{\pi}\log^{\frac{3}{2}}\Delta}$$

$\square$

Now, from here we get an upper bound on the number of subgraphs that connected l vertices at the top level. We bound it from above by counting combination of rooted subtrees, such the vertices are their root and the overall edges number is summed up to k. Thus:

$$\leq \zeta(\frac{3}{2})\binom{k-1}{l-1}\frac{\Delta^{2k}}{\sqrt{\pi}\log^{\frac{3}{2}}\Delta}$$

7 Tree Separator Lemma

8 Tanner - History Connected Equivalency

Imagine the following process, we apply t noisy iterations of the SWEEP rule, and then a single ideal (without noise) iteration. Denote by ξ a deviated configuration (can be either bits or checks), and by ξ' the result of applying an ideal SWEEP rule on ξ. Notice that if ξ is connected in the Tanner graph then ξ' is connected in the History graph **[COMMENT]** Require proof, For the classical case enumerate all the options can be done by hand. Thus:

$$\mathbf{Pr}\left[\xi \text{ connected in Tanner}\right] \leq \mathbf{Pr}\left[\xi' \text{ connected in Time} \mid \text{no noise at the } t+1 \text{ iteration}\right]$$

$$\leq \sum_{m}^{\infty} |\{ \text{ explanation to } \xi' \text{ over } m \text{ leaves restricted to no noise at the final iteration } \}| p^{\frac{m}{\Delta}}$$

$$\leq \sum_{m}^{\infty} |\{ \text{ explanation to } \xi' \text{ over } m \text{ leaves } \}| p^{\frac{m}{\Delta}}$$

$$\leq \mathbf{Pr}\left[\xi' \text{ connected in Time } \textbf{- Upper Bound} \right] \leq q^{\mathbf{Span}(\xi')} \leq q^{\mathbf{Span}(\xi)-1}$$

9 Intermediate Step Lemma

[COMMENT] The idea of this step is also to bound the case that all the bits are flipped at once, So checks are satisfied (and the argue above is not applied directly). Let ξ be a subset that connected component in the Tanner graph with a span $> \frac{d}{4}$ at time t, and let σ_i the event that at time i a connected with span at least $\sqrt{d}$ belongs to the error. Then:

$$\mathbf{Pr}\left[\xi\right] = \mathbf{Pr}\left[\xi \cap \cup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right]$$

So denote by τ the number of bits that require to be flipped in order to merge the flipped connected component to a one with a span greater than $d/4$, So τ has to be at least:

$$\tau\left(\sqrt{d}+1\right) \geq \frac{1}{4}d \Rightarrow \tau \geq \frac{1}{8}\sqrt{d}$$

Therefore:

$$\begin{aligned}\mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] &\leq \sum_{\xi^{(t-1)}} \mathbf{Pr}\left[\xi | \xi^{(t-1)}\right] \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq \sum_{\xi^{(t-1)}} p^{\frac{1}{8}\sqrt{d}} \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \leq p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

$$\begin{aligned}\mathbf{Pr}\left[\xi\right] &= \mathbf{Pr}\left[\xi \cap \cup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq t\mathbf{Pr}\left[\sigma_1\right] + p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

10 Configuration up to $C_z^{\perp}$